Automatic Number Plate Recognition System

Computer Vision-Based Vehicle Identification System

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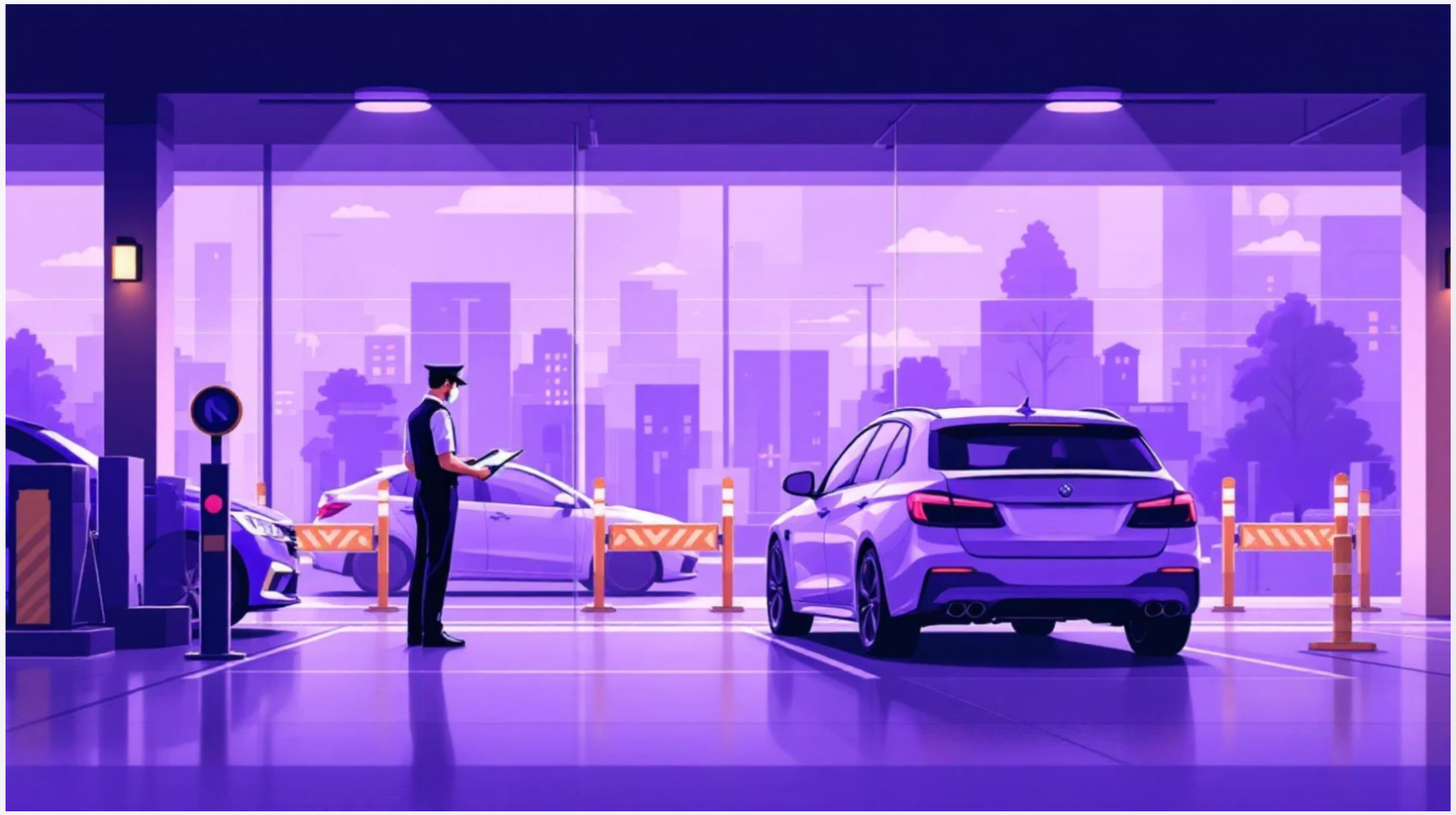
Problem Statement

Current Challenges

- Manual vehicle monitoring is inefficient and time-consuming
- High probability of human error in data entry
- Identification process lacks real-time automation
- Difficulty in scaling for high-traffic environments
- No centralized digital tracking system



Project Objectives

01

Automatic Plate Detection

Identify and locate vehicle number plates in images using computer vision techniques

02

Text Extraction

Extract alphanumeric characters from detected plates using Optical Character Recognition

03

Accuracy Enhancement

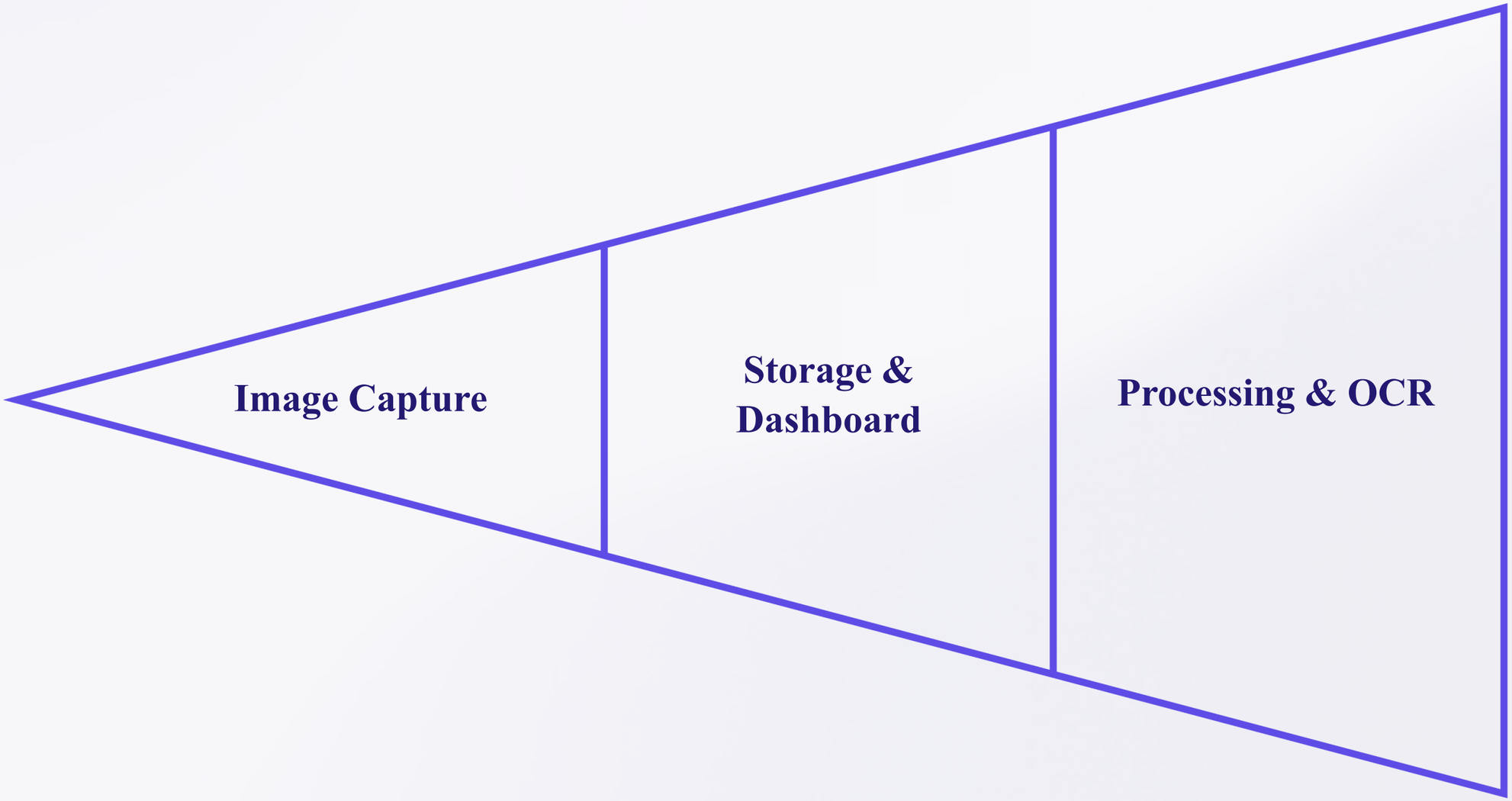
Improve recognition accuracy through image preprocessing and noise reduction

04

Data Management

Store extracted plate numbers with timestamps in structured database format

System Overview



Methodology

Image Capture

Acquire vehicle images via webcam or pre-existing image files

Preprocessing

Convert to grayscale, apply noise filtering, and enhance contrast

Edge Detection

Implement Canny edge detection algorithm to identify plate boundaries

Contour Detection

Locate potential plate regions through contour analysis and filtering

OCR Processing

Extract text using EasyOCR library with validation checks

Data Storage

Save plate numbers, timestamps, and metadata to CSV or SQLite database

Technologies Used

Python

Core programming language for system development

OpenCV

Computer vision library for image processing

EasyOCR

Optical character recognition engine

SQLite/CSV

Data storage and management

Streamlit

Web-based dashboard framework

NumPy

Numerical computing library

System Architecture

Modular Design

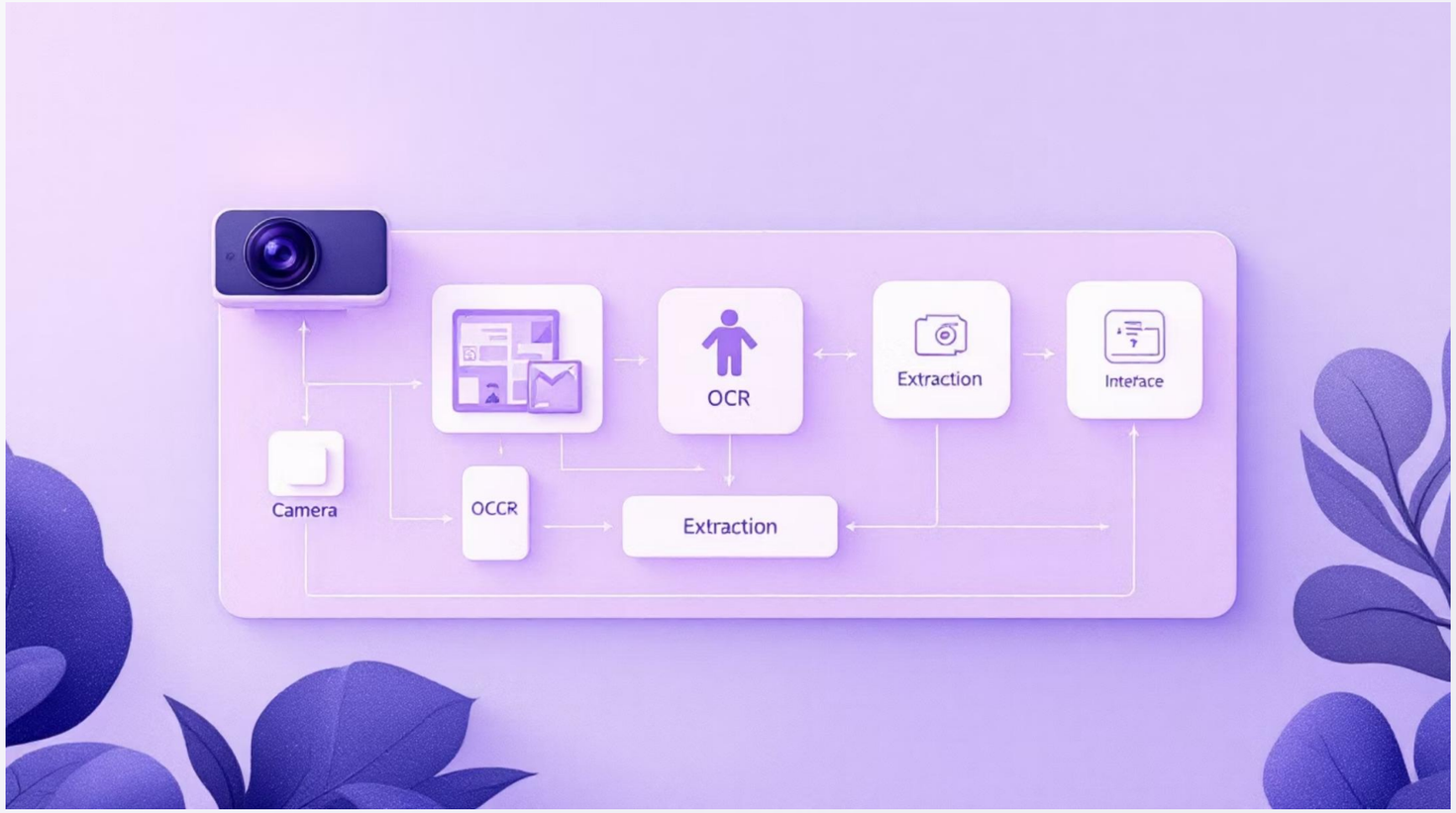
System employs separate modules for each processing stage, ensuring maintainability and scalability.

Real-Time Pipeline

Continuous processing from image capture through recognition to data storage with minimal latency.

Scalable Structure

Architecture supports expansion to multiple camera inputs and distributed processing nodes.



Results & Performance

Successful Detection

Accurately identifies number plates across various vehicle types and lighting conditions

Text Extraction

Extracts alphanumeric characters with good accuracy using OCR technology

Real-Time Processing

Operates continuously with webcam feed for live monitoring applications

Timestamp Storage

Records date, time, and plate information for audit trail and analysis

Challenges Encountered

Low Lighting Conditions

Poor illumination affects image quality and plate visibility, requiring adaptive preprocessing

Image Quality Issues

Blurred or noisy images from low-resolution cameras reduce recognition accuracy

OCR Limitations

Similar characters (0/O, 1/I) cause misreading, requiring validation logic

Perspective Problems

Angle and distance variations complicate plate localization and character extraction

Hardware Constraints

Limited processing power affects real-time performance with multiple camera feeds

Future Scope & Conclusion

Enhanced Detection Methods

- Implement YOLO (You Only Look Once) for faster, more accurate plate detection
- Integrate deep learning models for improved OCR accuracy
- Develop adaptive preprocessing for varying lighting conditions

System Expansion

- Enable real-time video analytics for multiple vehicles simultaneously
- Integrate cloud-based storage and processing infrastructure
- Develop mobile application for on-the-go monitoring

Conclusion

A functional, automated, and scalable ANPR system that reduces manual effort while maintaining accuracy. The modular architecture supports future enhancements and deployment in various security and traffic management applications.